

Multiple Choice

1. **Answer:** B

Options: A) cell-surface receptors; B) adhesion plaques; C) endosome formation; D) receptor-mediated endocytosis

Note: Correct option: B - adhesion plaques

2. **Answer:** B

Options: A) histones H 2 A, H 2 B, H 3, and H 4; B) linker DNA; C) histone H 2 A and H 2 B only; D) the nucleosome linker; E) histone H 3 only

Note: Correct option: B - linker DNA

3. **Answer:** B

Options: A) Frame shift mutations and nucleotide repeats; B) Base substitutions and small insertions and deletions; C) Base additions and large insertions; D) Silent mutations and missense mutations; E) Duplication mutations and triplet repeat expansions

Note: Correct option: B - Base substitutions and small insertions and deletions

4. **Answer:** A

Options: A) Temperature, humidity, and air currents; B) Plant species, root depth, and wind speed; C) Soil fertility, irrigation methods, and cloud cover; D) Water salinity, leaf color, and time of day; E) Leaf surface area, stem length, and moon phase

Note: Correct option: A - Temperature, humidity, and air currents

5. **Answer:** D

Options: A) Acquired characteristics are inherited through a blending of parental traits; B) Acquired characteristics are inherited but do not influence evolution; C) Acquired characteristics are inherited randomly, regardless of their use; D) Acquired characteristics cannot be inherited; E) Acquired characteristics are inherited based on use and disuse

Note: Correct option: D - Acquired characteristics cannot be inherited

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'A cell is the fundamental organizational unit of life.'

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Autotrophs and heterotrophs evolved simultaneously'.

Long Form Response

11. **Answer:** The reaction of the sensitive plant, *Mimosa pudica*, to touch demonstrates that transmission of impulses does indeed occur in plants.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** Fermentation. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key